UNIVERSITY OF MORATUWA

Second-Hand Vehicle Marketplace with Intelligent Search and Automated

Department of Computer Science and Engineering

SPRINT PLANNING DOCUMENT

Inventory Processing

Project ID (PID)	11
Group Number	25
Mentor	Mr. Bhanuka Siriwardhana
Methodology	Agile Scrum (2-week sprints, 3-member team)

Group Members

Registration No.	Name
230667F	Virusan T.
230670H	Vishula J.
230674A	R.P.M. Vithanage

1. Introduction

This document defines the Agile Scrum sprint plan for the development phase of the "Second-Hand Vehicle Marketplace with Intelligent Search and Automated Inventory Processing" project. It translates the high-level timeline from the Project Schedule (Gantt chart) and the scope defined in the Feasibility Study and Project Proposal into a series of executable two-week sprints, each with a clear sprint goal, backlog, and Definition of Done.

The team consists of three undergraduate developers who together act as the Scrum Team, sharing Product Owner and Scrum Master responsibilities on a rotating basis given the small team size.

2. Scrum Framework Overview

Given the team size (3 members) and academic time constraints, a lightweight Scrum process is adopted:

- Sprint length: 2 weeks.
- Total planned sprints: 7 (Sprint 1 – Sprint 7), aligned to the Project Schedule from Jun 26 to Oct 3.
- Sprint Planning: held on the first day of each sprint to select and estimate backlog items.
- Daily Stand-up: short (10–15 min) sync, 3 times a week minimum, given overlapping academic commitments.
- Sprint Review: demo of completed work at the end of each sprint, informally reviewed with the mentor where applicable.
- Sprint Retrospective: held after each review to identify improvements for the next sprint.
- Backlog tool: GitHub Projects / Issues is used to track the Product Backlog and Sprint Backlog, linked to version control.

2.1 Roles & Responsibilities

Member	**Primary Track**	**Secondary Track (cross-training)**	**Scrum Responsibilities**
Virusan T. (230667F)	Frontend & Requirements/Documentation	Backend (local API Gateway, Admin/Notification service) & DevOps (DB setup, CloudWatch, CI/CD)	Rotating Scrum Master, Sprints 1–2
Vishula J. (230670H)	Data & DevOps - ETL, Cloud & CI/CD	Shares embedding/search work with Backend track	Rotating Scrum Master, Sprints 6–7
R.P.M. Vithanage (230674A)	Backend - Auth, Marketplace & Search services	Shares embedding/search & CI/CD tasks with DevOps track	Rotating Scrum Master, Sprints 3–5

Every member owns at least one Frontend, one Backend, and one DevOps/Infra task across the plan, so the workload is balanced by complexity rather than strictly by track - this also builds cross-functional familiarity with the full stack, which is valuable given the team size.

2.2 Definition of Done (applies to all sprints)

- Code is committed to the shared GitHub repository and passes the GitHub Actions CI checks (lint + tests).
- Feature has associated unit tests (where applicable) with a passing status.
- Feature has been manually verified using Postman or the local Docker environment.
- Relevant documentation (README, API notes, or design doc) is updated.
- Code has been reviewed by at least one other team member before merging to the main branch.
- No critical or high-severity bugs remain open against the item.

3. Product Backlog (Epics)

The Product Backlog is organised into epics derived directly from the project scope. Individual sprint backlogs below break these epics into sprint-sized tasks.

Epic	Description
EPIC-1: Documentation & Planning	Proposal, feasibility study, schedule, SRS, design document, test plan
EPIC-2: Auth & User Service	Registration, login, JWT authentication, role-based access
EPIC-3: Marketplace & Search Service	Dealer profiles, listings, favourites, filter search, NL/semantic search
EPIC-4: Ingestion / ETL Service	Bulk CSV/JSON/ZIP upload, validation, normalization, image processing, embeddings
EPIC-5: Admin & Notification Service	Admin monitoring, audit logs, upload status, SES email notifications
EPIC-6: Frontend Application	Dealer, buyer, admin, and guest interfaces; API Gateway integration
EPIC-7: Cloud-Native Deployment	Terraform IaC, API Gateway config, CI/CD pipeline via GitHub Actions
EPIC-8: Testing & Quality Assurance	Unit, component, integration testing, and UAT

4. Sprint Plan Overview

The table below maps each sprint to the Project Schedule (Gantt chart) timeline.

Sprint	Duration	Theme
Sprint 1	Jun 26 – Jul 9	Project initiation & core documentation
Sprint 2	Jul 10 – Jul 23	Requirements, design & environment setup
Sprint 3	Jul 24 – Aug 6	Auth service, API gateway & marketplace core
Sprint 4	Aug 7 – Aug 20	Ingestion pipeline: upload & queueing
Sprint 5	Aug 21 – Sep 3	ETL processing, admin service & frontend kickoff
Sprint 6	Sep 4 – Sep 17	Frontend integration & cloud deployment
Sprint 7	Sep 18 – Oct 3	Testing, UAT, demo & final report

5. Sprint Backlogs

Sprint 1 (done)

Jun 26 – Jul 9, 2026

Sprint Goal: Complete the project proposal and feasibility study, and establish the team's shared understanding of scope, architecture, and schedule.

Task	Assignee	Pts	Priority	status
Draft & finalize Project Proposal document	All	5	High	done
Draft & finalize Feasibility Study (financial, technical, resource, risk)	All	8	High	done
Build Project Schedule / Gantt chart	Virusan T.	3	High	done
Set up GitHub repository, branching strategy & board	R.P.M. Vithanage	2	Medium	done
Research & shortlist technology stack (React, NestJS, AWS services)	Vishula J.	3	Medium	done

Sprint 2

Jul 10 – Jul 23, 2026

Sprint Goal: Produce the Requirements/SRS and Design documents, and stand up the local backend environment and databases.

Task	Assignee	Pts	Priority	status
Requirement Gathering & SRS document	All	8	High	done
Design Document (ER diagram, architecture diagram, API contracts)	All	8	High	done
Initialize GitHub repo structure for microservices & Docker Compose setup	R.P.M. Vithanage	3	High	done
Set up local PostgreSQL with pgvector & pg_trgm extensions	Vishula J.	3	High	done
Scaffold NestJS Auth & User Service module	R.P.M. Vithanage	5	High	done
Set up React project skeleton & UI component library	Virusan T.	3	Medium	done

Sprint 3

Jul 24 – Aug 6, 2026

Sprint Goal: Deliver a working Auth service (including tiered dealer verification) and Local API Gateway, and begin the Marketplace service with hybrid search.

Task	Assignee	Pts	Priority	status
Implement registration, login & JWT authentication	R.P.M. Vithanage	8	High	Done
Implement role-based access control (dealer / buyer / admin)	R.P.M. Vithanage	5	High	done
Implement dealer_type selection (Individual/Business), document upload capture, and refresh token issuance/exchange/revocation	R.P.M. Vithanage	5	High	done

Set up local API Gateway routing to services	Rashmi	5	High	
Develop Marketplace Service: dealer profiles & vehicle listing CRUD	Virusan T. + rashmi	8	High	done
Implement filter-based search (make, model, price, mileage, fuel type)	Vishula J.	5	Medium	done
Implement intelligent vehicle search pipeline with query parsing, Groq fallback, pg_trgm keyword search and pgvector semantic search	Vishula J.	8	High	partial
Unit tests for Auth & Marketplace modules	Rashmi and vishula	3	Medium	done

Sprint 4

Aug 7 – Aug 20, 2026

Sprint Goal: Build the ingestion path of the ETL pipeline: bulk file upload to S3 and asynchronous job queueing via SQS.

Task	Assignee	Pts	Priority	Status
Implement CSV/JSON/ZIP upload endpoint to Amazon S3	Vishula J.	8	High	ongoing
Configure SQS queue & job status tracking model	Rashmi	5	High	ongoing
Implement upload job status API for dealer dashboard	Virusan T.	3	Medium	ongoing
Build synthetic vehicle data generation tool for test CSV/JSON files	Virusan T.	5	Medium	ongoing
Continue Marketplace favourites & buyer-facing endpoints	R.P.M. Vithanage	5	Medium	ongoing
Unit testing for ingestion endpoints (ongoing)	All	3	Medium	ongoing

==========before mid evaluation ==========

Sprint 5

Aug 21 – Sep 3, 2026

Sprint Goal: Complete the ETL transform/load stages (validation, image processing, embeddings), stand up the Admin & Notification service, and kick off frontend interface development - split between both frontend contributors.

Task	Assignee	Pts	Priority
Implement record validation & normalization rules in ETL worker (incl. vehicle_dictionaries lookups)	Vishula J.	8	High
Implement image-to-listing mapping, resizing & compression (Sharp)	R.P.M. Vithanage	8	High

Task	Assignee	Pts	Priority
Generate & store search metadata / embeddings for uploaded listings	Vishula J. / R.P.M. Vithanage	5	High
Admin & Notification Service: audit logs & upload status reporting	Virusan T.	5	Medium
Admin: pending dealer document review & tiered capability gating (listing cap for Individual, bulk-upload gate for Business)	Virusan T.	5	Medium
Integrate AWS SES for upload success/failure email notifications	Virusan T.	3	Medium
Build Buyer & Guest UI screens (browse, search, detail, favourites)	Virusan T.	4	High
Build Dealer & Admin UI screens (upload, dashboard, listings mgmt, user mgmt)	R.P.M. Vithanage	4	High
Component testing for ETL & Admin modules	All	3	Medium

v1.2: "Build Dealer, Buyer & Admin UI screens" (previously 8 pts solo on R.P.M. Vithanage) is now split into a Buyer/Guest half (Virusan T.) and a Dealer/Admin half (R.P.M. Vithanage). Admin document review is a new task, added to Virusan T.'s existing Admin/Notification work.

Sprint 6

Sep 4 – Sep 17, 2026

Sprint Goal: Connect the frontend to backend APIs and deploy the full system to AWS using Infrastructure-as-Code and CI/CD - frontend integration shared across both contributors.

Task	Assignee	Pts	Priority
Connect Buyer & Guest-facing screens (search, detail, favourites) to API Gateway	Virusan T.	4	High
Connect Dealer & Admin-facing screens (upload, dashboard, user mgmt, document review) to API Gateway	R.P.M. Vithanage	4	High
Handle upload progress, error states & notifications in UI	R.P.M. Vithanage	5	Medium
Write Terraform scripts for RDS, S3, Lambda, SQS, ECR & SES	Vishula J.	8	High
Configure production AWS API Gateway & IAM roles per service	Vishula J.	5	High
Set up CI/CD pipeline: GitHub Actions build, ECR push & Lambda deploy	All	8	High
Set up CloudWatch logging & billing alerts	Virusan T.	3	Medium

v1.2: "Connect frontend screens to API Gateway" (previously 8 pts solo on Virusan T.) is now split Buyer/Guest (Virusan T.) vs Dealer/Admin (R.P.M. Vithanage). "Handle upload progress/error UI" moved from Virusan T. to R.P.M. Vithanage, since it's dealer-facing and now pairs with his Dealer/Admin integration work - this is what actually fixes the "frontend alone" problem: R.P.M. Vithanage had zero tasks in this sprint before.

Sprint 7

Sep 18 – Oct 3, 2026

Sprint Goal: Verify system quality end-to-end, complete documentation, and deliver the final submission package.

Task	Assignee	Pts	Priority
Prepare & execute Test Plan Report	All	5	High
Integration testing across all microservices	All	8	High
User Acceptance Testing (UAT) with sample dealer/buyer scenarios	All	5	High
Fix defects found during integration testing & UAT	All	5	High
Record, edit & submit demo video and YouTube link	Virusan T.	3	Medium
Compile Final Report & README.txt	All	5	High

6. Sprint Velocity & Capacity Notes

With 3 team members working part-time alongside academic coursework, an average team velocity of approximately 25–30 story points per 2-week sprint is assumed for planning purposes. Story point estimates above should be re-calibrated after Sprint 1 and Sprint 2 once actual velocity is observed, and the Sprint 4–7 backlogs may be re-prioritised at the end of each Sprint Retrospective.

6.1 Workload Balance Across the Team

Recalculated for v1.2 to reflect the Sprint 5/6 frontend split and the two new tasks. Vishula J.'s tasks are unchanged from v1.1.

Member	Frontend Pts	Backend Pts	DevOps Pts	Docs & Testing Pts	Approx. Total
Virusan T.	11	30	13.7	25.3	~72
R.P.M. Vithanage	13	30.5	7.7	19.3	~78.5
Vishula J. (unchanged)	0	17	42.2	22.3	~81.5

6.2 Other Gaps Worth Discussing (not yet added to the backlog)

These fall inside Vishula J.'s existing DevOps/ETL tasks rather than being added as new line items here, since her assigned tasks were left unchanged - but they aren't currently explicit anywhere in the plan, so the team should confirm they're covered:

- vehicle_dictionaries reference table (canonical makes/models/fuel/transmission, with alias promotion) - implicitly needed by Sprint 3's search pipeline and Sprint 5's normalization task, but not called out as its own deliverable.
- RDS Proxy in front of the RDS instance - needed so the ETL Load stage's concurrent invocations (Map State, MaxConcurrency 10) don't exhaust the database connection pool; a natural fit inside Sprint 6's Terraform task.

- Internal API Gateway/ALB for service-to-service calls (e.g. Admin approving a dealer) - distinct from the public API Gateway task already in Sprint 3/6; worth confirming it's in scope for Sprint 6's "production AWS API Gateway" task or needs its own line.

7. Risks Carried into Sprint Execution

- Poor quality uploaded data - mitigated via validation tasks scheduled in Sprint 5.
- ETL / processing failure - mitigated via SQS retry logic built in Sprint 4.
- Cloud cost overrun - monitored via CloudWatch billing alerts set up in Sprint 6.
- Team member unavailability - mitigated by shared documentation and GitHub-tracked tasks so any member can pick up pending work.
- Deployment/integration errors - addressed through incremental local Docker testing before each cloud deployment step in Sprint 6.

8. Next Steps

This sprint plan should be treated as a living document. At the end of each Sprint Review and Retrospective, the team will update completed story points, carry over unfinished items to the next sprint, and adjust upcoming sprint backlogs to reflect actual progress against the Project Schedule.